

Two Ways to Find a Compound Figure's Area

ELA + MATH • WORD STUDY

Mr. H Math · Day 7

WRITE YOUR NAME — one letter per box, straight and neat

[illegible]

Date: _____

Score: ____ / 10

Yesterday you split an L-shaped floor into two rectangles and added the pieces. Today you get a **second tool** for the same job — and a real choice between them. Every blank you fill in unlocks a word we'll use for the rest of the Area unit, so read closely and choose your words carefully!

Section A · Guess the Word

Fill in the blanks to find the word that means "break a shape into smaller pieces."

D E C M P O E

Write the two missing letters in the empty boxes.

To _____ a compound figure means to break it into rectangles you already know how to find the area of.

Every rectangle you cut it into must fit perfectly — no gaps, no overlaps.

Section B · Meet the Compound Figure Family

Use the word bank to complete each sentence.

compound figure

decompose

enclose

subtract

- 1 A shape made of two or more rectangles put together — like an L or a T — is called a _____.
Example: an L-shaped floor, a T-shaped room.
- 2 One way to find its area is to _____ it into rectangles, then add each piece.
Example: split an L into 8×3 and 4×2 , then add: $24 + 8 = 32$.
- 3 Another way is to _____ the whole shape inside one big rectangle, then _____ the extra piece that isn't really there.
Example: draw a 10×6 rectangle around the shape, then take away the 3×2 corner: $60 - 6 = 54$.

Section C · Try It: Add the Pieces, or Enclose and Subtract

Study the worked example, then solve your own using either method.

Worked Example · Add the Pieces

$$8 \times 3 + 4 \times 2 = 24 + 8 = 32 \text{ sq ft}$$

↑ piece A

↑ piece B

↑ total area

Your turn: A room is shaped like a rectangle with a rectangular corner missing. The big rectangle is $10\text{ ft} \times 7\text{ ft}$, and the missing corner is $4\text{ ft} \times 3\text{ ft}$. Draw the shape in the grid, then write the subtraction sentence and label the big rectangle, the cut-out, and the final area.

[illegible]

$$70 \text{ sq ft} - 12 \text{ sq ft} = 58 \text{ sq ft}$$

EXPLORE · SUMMARY

Compound figures don't stop at straight corners. Sometimes a corner is sliced off on a **diagonal** instead of straight across, and that piece becomes a triangle instead of a rectangle. A triangle needs a new trick — cutting it in half from a rectangle you already know — and that's exactly where the Area unit heads next.

☆ Add the pieces, or subtract the extra — both roads lead to the same area! ☆

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Word Study Meets Number Sense · Day 7 wksht · Prepares students for Area of Triangles (6.G.A.1)

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